Islamabad AQI Predictor

End-to-End Air Quality Index Prediction System

Final Project Report June 2026

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GitHub	github.com/maham-bhatti/aqi-predictor
Live App	aqi-predictor-natgjjfhdmgmvne6pkd9op.streamlit.app
Data Sources	AQICN API · Open-Meteo API · Open-Meteo Archive
Feature Store	Supabase (PostgreSQL)
CI/CD	GitHub Actions (hourly feature + daily training)

1. Executive Summary

This report documents the complete end-to-end development of the Islamabad AQI Predictor a fully serverless, automated machine learning system that forecasts Islamabad's Air Quality Index (AQI) for the next 24, 48, and 72 hours. The system is built on a modern MLOps stack and deployed as an interactive real-time web dashboard.

The project fulfils all four deliverables defined in the project specification:

- An automated hourly feature pipeline that fetches live pollution and weather data
- A daily training pipeline that retrains ML models on growing historical data
- An interactive Streamlit dashboard with real-time AQI, 3-day forecasts, historical trends, and health alerts
- This detailed report documenting every component of the system

The best-performing model is Ridge Regression, achieving an RMSE of 10.94 and R^2 of 0.363 on the 24-hour horizon. Model performance degrades at longer horizons (48h, 72h), which is expected given the inherent volatility of air quality data and limited training history. As more hourly data accumulates automatically, model accuracy will improve over time.

2. System Architecture

The system follows a four-stage serverless MLOps pipeline as illustrated below. All stages run automatically with zero manual intervention after initial deployment.

2.1 Pipeline Overview

#	Stage	Description	Schedule / Trigger
1	**Feature Pipeline**	Fetches AQI + weather, computes features, stores in Supabase	Every hour (GitHub Actions cron)
2	**Backfill Script**	One-time backfill of 365 days of historical data	Manual (one-time run)
3	**Training Pipeline**	Fetches features from Supabase, trains 5 models, saves best	Every day at 02:00 UTC (GitHub Actions)
4	**Streamlit Dashboard**	Loads live data + model, shows predictions and charts	Always-on (Streamlit Cloud)

2.2 Technology Stack

Layer	Technology	Purpose
Data Ingestion	AQICN API + Open-Meteo API	Real-time AQI, pollutants, weather
Historical Data	Open-Meteo Archive API	365-day backfill for training

Feature Store	Supabase (PostgreSQL)	Hourly features + target storage
ML Models	scikit-learn, XGBoost	Ridge, RF, GBM, XGBoost, Linear
CI/CD	GitHub Actions	Automated hourly + daily pipeline runs
Dashboard	Streamlit + Plotly	Interactive web dashboard
Deployment	Streamlit Cloud	Free serverless hosting
Version Control	GitHub	Code + model artifacts storage

3. Data Collection & Feature Engineering

3.1 Data Sources

Two primary APIs provide all raw data for the system:

- **AQICN API:** AQICN API, Real-time AQI and individual pollutant readings (PM2.5, PM10, NO_2 , O_3 , CO, SO_2) for the Islamabad monitoring station. Updated hourly.
- **Open-Meteo Forecast:** Open-Meteo API, Real-time and forecast weather data: temperature (°C), relative humidity (%), wind speed (km/h), surface pressure (hPa), and precipitation (mm).
- **Open-Meteo Archive:** Open-Meteo Archive API, Historical hourly air quality and weather data used for the backfill, covering 365 days from June 2025 to June 2026.

3.2 Feature Engineering

The feature pipeline computes 24 features from raw data for each hourly record:

Category	Features	Description
Pollutants (7)	`aqi, pm25, pm10, no2, o3, co, so2`	Raw pollutant readings from AQICN API
Weather (5)	`temperature, humidity, windspeed, pressure, precipitation`	Current weather conditions from Open-Meteo
Time (4)	`hour, day_of_week, month, is_weekend`	Cyclical time-based features capturing daily/weekly patterns
Lag Features (4)	`aqi_lag_1h, _3h, _6h, _24h`	AQI value from 1, 3, 6, and 24 hours ago
Rolling Averages (3)	`aqi_rolling_3h, _6h, _24h`	Mean AQI over past 3, 6, 24 hour windows
Change Rate (1)	`aqi_change_rate`	AQI change from previous hour (trend indicator)

3.3 Target Variables

Three target variables are computed for supervised learning:

- aqi_next_24h, AQI value 24 hours ahead (primary prediction target)
- aqi_next_48h, AQI value 48 hours ahead
- aqi_next_72h, AQI value 72 hours ahead

3.4 Dataset Statistics

- Training records: 6,975 hourly rows
- Test records: 1,744 hourly rows
- Total dataset: 8,719 hourly records (~363 days)
- Date range: June 2025 - June 2026 (covers all four seasons)
- Storage: Supabase PostgreSQL with upsert on timestamp (no duplicates)

4. Model Training & Evaluation

4.1 Models Evaluated

Five regression models were trained and evaluated for each forecast horizon (24h, 48h, 72h). All models predict the target AQI value directly as a continuous variable.

4.2 Model Performance - 24h Horizon

Model	RMSE	MAE	R^2
Linear Regression	10.94	8.09	**0.363**
Ridge Regression ★	10.94	8.09	**0.363**
Random Forest	12.39	8.87	**0.182**
Gradient Boosting	12.16	8.65	**0.212**
XGBoost	12.10	8.67	**0.220**

★ Ridge Regression selected as best model (tied R^2 with Linear Regression but more robust against overfitting due to L2 regularization).

4.3 Multi-Horizon Performance

Horizon	RMSE	MAE	R^2
24h Forecast	10.94	8.09	**0.363**
48h Forecast	13.68	10.24	**0.005**
72h Forecast	16.90	11.86	**-0.098**

Performance degrades at 48h and 72h horizons. This is expected behavior for AQI forecasting: air quality is highly non-linear and depends on meteorological events (dust storms, wind direction shifts) that linear models cannot capture beyond 24 hours. The models will improve as: (1) more training data accumulates from the automated daily pipeline, and (2) future iterations add meteorological forecast features as inputs.

4.4 Why Ridge Regression Outperformed Tree Models

Islamabad AQI data (June 2025–2026) shows a relatively stable baseline with periodic spikes. Ridge Regression captures the dominant linear relationship between current AQI and lagged AQI values effectively. Tree-based models (Random Forest, XGBoost) would need significantly more data and hyperparameter tuning to generalize better.

5. Automated Pipelines (CI/CD)

5.1 Feature Pipeline - Runs Every Hour

File: feature_pipeline/feature_pipeline.py | Workflow: .github/workflows/feature_pipeline.yml

Steps executed each hour:

1. Fetch current AQI data from AQICN API for Islamabad
2. Fetch current weather from Open-Meteo (temperature, humidity, wind, pressure, precipitation)
3. Retrieve last 24 hourly records from Supabase to compute lag and rolling features
4. Combine all features into a single record and upsert to Supabase (on_conflict=timestamp prevents duplicates)

5.2 Training Pipeline - Runs Every Day at 02:00 UTC

File: training_pipeline/training_pipeline.py | Workflow: .github/workflows/training_pipeline.yml

Steps executed each day:

5. Fetch all historical records from Supabase where targets are not null
6. Split data 80/20 chronologically (time-series split, no random shuffling)
7. Train and evaluate 5 models (Linear, Ridge, Random Forest, Gradient Boosting, XGBoost) for each of 3 horizons
8. Save best model (by RMSE on 24h) + scaler + features + metrics to models/ directory
9. Commit updated model files back to GitHub (auto-commit via git in the workflow)

5.3 Backfill Script

File: training_pipeline/backfill.py, run once to populate Supabase with 365 days of historical data from the Open-Meteo Archive API. Generates lag features, rolling averages, and forward-fills target columns (aqi_next_24h, _48h, _72h).

6. Streamlit Dashboard

6.1 Dashboard Components

The dashboard at aqi-predictor-natgjjfhdmgmvne6pkd9op.streamlit.app contains the following sections:

- **Current AQI:** Real-time AQI gauge with color-coded severity and delta showing 24h forecast direction
- **Weather Panel:** Live weather metrics: temperature, humidity, wind speed, pressure
- **Pollutant Chart:** Bar chart of individual pollutant levels (PM2.5, PM10, NO_2, O_3, CO, SO_2)
- **3-Day Forecast:** 3 forecast cards (24h, 48h, 72h) with AQI value, category badge, and health advice
- **AQI History:** 30-day AQI time series with Good / Moderate / Unhealthy threshold lines
- **Weather Forecast:** 24-hour temperature and rainfall probability dual-axis chart
- **AQI Heatmap:** AQI heatmap by hour-of-day vs date showing daily/weekly pollution patterns
- **Model Metrics:** Model R^2 and RMSE scores for all three forecast horizons
- **AQI Reference Scale:** Colour-coded scale from Good (0–50) to Hazardous (300+)
- **Health Alerts:** Dynamic banners: green for good air, orange for caution, red for health warning

6.2 Technology Details

- Streamlit for layout and interactivity
- Plotly for all charts (gauge, bar, scatter, heatmap, dual-axis)
- Custom CSS with Inter + Space Grotesk fonts, light blue gradient theme
- Data cached with @st.cache_data(ttl=3600), refreshes every hour
- Models loaded with @st.cache_resource, loaded once per session

7. Repository Structure

GitHub: github.com/maham-bhatti/aqi-predictor

Path	Description
app/streamlit_app.py	Main Streamlit dashboard application
feature_pipeline/feature_pipeline.py	Hourly feature collection and storage
training_pipeline/training_pipeline.py	Daily model training script
training_pipeline/backfill.py	One-time historical data backfill
training_pipeline/notebooks/eda_analysis.ipynb	Exploratory Data Analysis notebook
models/model_24h.pkl	Trained Ridge Regression model (24h)
models/model_48h.pkl	Trained Ridge Regression model (48h)
models/model_72h.pkl	Trained Ridge Regression model (72h)
models/scaler.pkl	StandardScaler fitted on training data
models/features.pkl	List of feature names (24 features)
models/metrics.json	Model performance metrics + feature list
.github/workflows/feature_pipeline.yml	GitHub Actions: runs feature_pipeline.py hourly
.github/workflows/training_pipeline.yml	GitHub Actions: runs training_pipeline.py daily
requirements.txt	Python dependencies
.env	Environment variables (not committed)

8. Key Findings from EDA

- Islamabad AQI is highly seasonal, significantly worse in winter months (Nov–Feb) due to temperature inversions trapping pollutants, and better during monsoon (Jul–Sep) when rain washes pollutants away.
- Strong diurnal cycle: AQI peaks in morning (7–9 AM) and evening rush hours (6–9 PM), dips during midday when convection disperses pollutants.
- PM2.5 is the dominant pollutant, it is the primary driver of AQI values for Islamabad. Other pollutants (NO_2 , O_3 , SO_2) are reported inconsistently by the monitoring station.
- Wind speed is inversely correlated with AQI, higher winds disperse pollutants more effectively. Days with wind speed > 15 km/h consistently show AQI < 80.
- Lag features are the strongest predictors, aqi_lag_1h and aqi_rolling_3h have the highest correlation with future AQI values, confirming the persistent nature of air pollution.
- June 2026 shows a sudden spike to AQI 150+ consistent with pre-monsoon dust storms common in the region during May–June.

9. Limitations & Future Work

9.1 Current Limitations

- Ridge Regression R^2 of 0.36 for 24h is moderate, the model explains about 36% of AQI variance, which is acceptable for a first iteration but leaves room for improvement.
- 48h and 72h models have near-zero R^2, predictions at these horizons are essentially the historical mean, not meaningful forecasts.
- Individual pollutant data (PM10, NO_2, O_3, SO_2) is frequently missing from the AQICN station, only PM2.5 is consistently reported.
- Only 8 features are actually used by the trained model (features_in_=8) vs 24 features defined indicates the training script uses a subset. This should be corrected in the next training run.

9.2 Planned Improvements

- Add meteorological forecast features (next 24h wind, temperature, humidity from Open-Meteo) as model inputs to improve 48h/72h accuracy
- Implement LSTM or GRU time-series model for better long-horizon forecasting
- Add SHAP feature importance explanations to the dashboard
- Fix training pipeline to use all 24 features consistently
- Add model drift detection and automatic retraining alerts

10. Submission Checklist

#	Requirement	Status
1	End-to-end AQI prediction system	✓ Complete
2	Scalable, automated pipeline (GitHub Actions)	✓ Complete
3	Interactive dashboard with real-time + forecast data	✓ Complete
4	Detailed report documenting everything achieved	✓ Complete
5	Feature pipeline (hourly, GitHub Actions)	✓ Complete
6	Backfill historical data (365 days)	✓ Complete
7	Training pipeline (daily, GitHub Actions)	✓ Complete
8	Multiple ML models evaluated (5 models)	✓ Complete
9	RMSE, MAE, R^2 evaluation metrics	✓ Complete
10	EDA notebook	✓ Complete
11	Health alerts for hazardous AQI	✓ Complete
12	GitHub repository	✓ github.com/maham-bhatti/aqi-predictor

Appendix - **Environment Variables**

The following environment variables must be set (in .env locally, and GitHub Secrets for CI/CD):

Variable	Purpose	Where to Get It
`AQICN_TOKEN`	Real-time AQI API key	56bd3e6bf4e43f63b9ee23ba692a43bf49da9767
`SUPABASE_URL`	Supabase project URL	https://zejndpznhjrjejvubsnq.supabase.co
`SUPABASE_KEY`	Supabase anon/service key	eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6Inplam5kcHpuaGpyamVqdnVic25xIiwicm9sZSI6InNlcnZpY2Vfcm9sZSIsImlhdCI6MTc3OTg1MTgzNCwiZXhwIjoyMDk1NDI3ODM0fQ.PUPugZqPODnVrdCmfiLOMSrFQ6T3xi146IBaKGMdKdM

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